

Master Prince

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WORK EXPERIENCE

Full Stack AI Developer

May 2025 – June 2026

M37 Labs

Delhi NCR

- Shipped **5+ production LLM agentic systems** (planning, **tool/function calling**, routing, persistent state) for enterprise clients, with a routing layer spanning **10+ agents** on **FastAPI + Next.js** at **85% uptime**
- Engineered **end-to-end RAG pipelines** — ingestion, chunking, **embeddings**, **Qdrant/ChromaDB** indexing, latency-optimized retrieval — incl. a **LangChain SQL** chatbot over **200K+ publications**
- Implemented **MCP servers** with AES-256 encrypted OAuth storage; routed **GPT-4, Claude, Gemini** via **LiteLLM** with provider fallbacks for cost/quality control
- Developed a **layout-aware document AI pipeline** — PyMuPDF/pdfplumber parsing with **Tesseract OCR** fallback for scanned PDFs — feeding a hybrid **spaCy** scoring model that reduced manual screening time by **70%**; also shipped a **scikit-learn** email classifier (**95%+ accuracy**)
- Slashed API latency **40%** and DB load **60%** via **Redis** caching over **PostgreSQL**; deployed with **Docker** on **AWS EC2**

Full Stack Developer Intern

Dec 2024 – Feb 2025

Luchkee Health Pvt. Ltd.

Remote

- Architected a MERN healthcare platform with **JWT auth, RBAC**, and encrypted patient data storage; reduced **MongoDB** query time by **30%** and dashboard load from 3s to under **1s** via query/index optimization

TECHNICAL SKILLS

LLM & Agents: OpenAI, Claude, Gemini, Llama, Qwen; LangChain, LiteLLM, **MCP**, tool calling, ReAct agents, prompt engineering, guardrails

RAG & Document AI: Qdrant, ChromaDB, embeddings, chunking, retrieval optimization; **OCR** (Tesseract, EasyOCR), layout-aware PDF parsing (PyMuPDF, pdfplumber)

Fine-tuning & Local Inference: LoRA, QLoRA, MLX-LM, Ollama, quantized models, self-hosted / on-prem deployment

ML & Time-Series: scikit-learn, XGBoost, spaCy, Pandas, NumPy, time-series forecasting

Backend & DevOps: Python, FastAPI, TypeScript, Next.js, REST, WebSockets/SSE, PostgreSQL, Redis, MongoDB, Docker, AWS, Git, CI/CD

KEY PROJECTS

Semantic Memory Engine | Python, RAG, QLoRA, ChromaDB, Ollama, LLM Evaluation

May 2026 | [GitHub](#)

- Designed a memory + prompt-compression engine for **local LLMs**: **~10x token compression** via symbolic context encoding, with **ChromaDB** retrieval, deduplication, and token-budgeted assembly
- QLoRA fine-tuned Qwen2.5-3B** to reason over the compressed format; **evaluation harness** showed **-26% hallucination rate** and +0.61 answer relevance vs. base model

Mental Health AI — Self-Hosted LLM API | Python, MLX-LM, LoRA, FastAPI

Mar 2026 | [GitHub](#)

- Trained two **Qwen2.5-3B** models fully **on-device** (LoRA/MLX): a 14-class classifier (12,880 examples) and a response generator — **zero cloud dependency** for a safety-sensitive domain
- Served both behind a **FastAPI** REST API with multi-turn session memory, batch classification, and structured logging

SiteBOT — Agentic Chatbot Platform | Python, FastAPI, ReAct Agents, Redis, Next.js

May 2026 | [GitHub](#)

- Created a multi-turn **ReAct agent** that plans and executes **tool calls** (booking, lead capture, FAQ search, human handoff) with configurable **guardrails**, streamed over **SSE**
- Persistent **RAG memory** with **Redis + PostgreSQL** caching and automated compression; multilingual low-latency voice mode (STT/TTS)

PM2.5 Time-Series Forecasting — Delhi CPCB Sensor Data | Python, scikit-learn, XGBoost, NumPy

May 2026 | [GitHub](#)

- Forecasted PM2.5 from **6 years of CPCB sensor time-series data**; **Random Forest / XGBoost** reached **$R^2 = 0.95$** (MAE 12.4), plus a NumPy neural network built from scratch

EDUCATION

B.Tech in Computer Science & Engineering (AI)

Aug 2021 – May 2025

Jamia Hamdard University, New Delhi

CGPA: 7.2